Final Paper

Linda Herrera, Natalie Chen, June McNally

Introduction

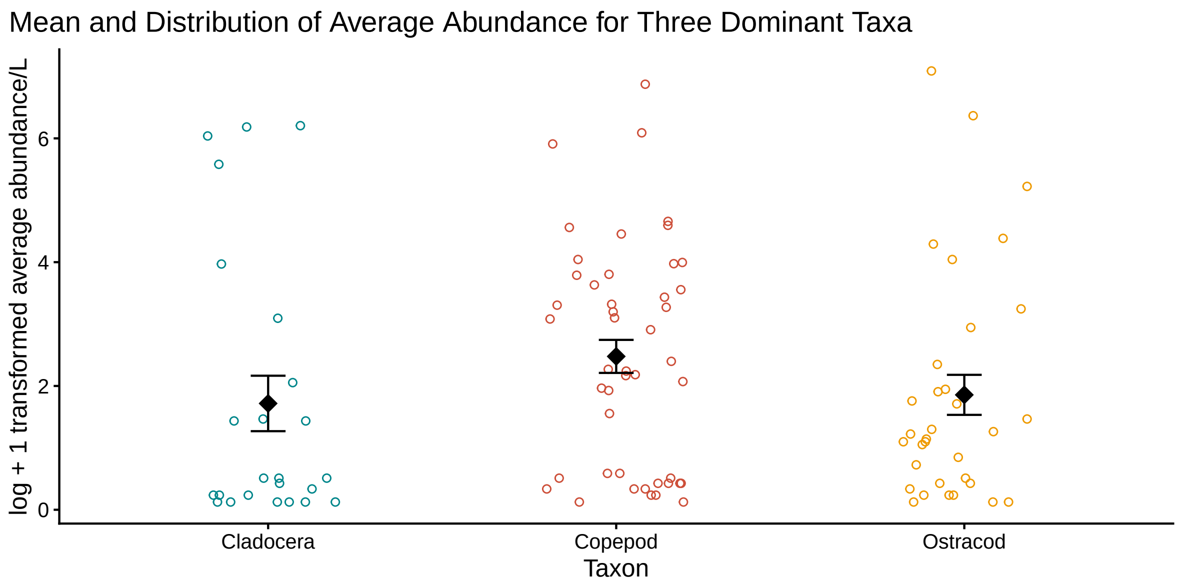
- The North Campus Open Space (NCOS) in Goleta is a restored wetland managed by the UCSB Cheadle Center. From 1966 to 2013, the site operated as a 9-hole golf course. Restoration began in 2017, and aquatic invertebrate monitoring began shortly after that in 2018. Aquatic invertebrates are small, spineless creatures that play a critical role in wetland food webs, as they are a primary link between producers and higher consumers such as waterbirds and fish. Their sensitivity to environmental stressors also makes them valuable indicator species for assessing restoration progress both at NCOS and globally. In restored wetlands ranging from New Zealand (Suren, Lambert, and Sorrell 2011) to the southeastern United States (Batzer et al. 2015) to Northeastern China (Lu et al. 2020), invertebrate assemblages have consistently tracked the trajectory of ecological recovery, with community composition and richness responding measurably to improvements in habitat conditions. Studies of zooplankton in restored Wisconsin wetlands similarly found that taxon richness correlated strongly with time since restoration, taking approximately six to seven years to resemble the least impacted reference sites (Dodson and Lillie 2001).
- This is important to understanding the significance of invertebrates at NCOS; they directly show the progress of the restoration methods just as they have been around the world. Temperature is one strong abiotic driver of the physical processes in aquatic ecosystems and wetlands, which can be assessed using the abundance of aquatic invertebrates. Each species requires a specific temperature range for optimal performance; when approaching their thermal limits, organisms show signs of stress, resulting in changes in behavior resulting in migration or death (Bonacina et al. 2022). Each of the taxa have specific ideal temperature ranges and generally disappear at more extreme temperatures, with all ideal temperatures being around a range of 18-30℃ (Li et al. 2009).
- When the invertebrates function at their ideal temperature, it leads to an overall healthier aquatic ecosystem. These relationships led to the main question of our study: what is the relationship between the abundance of aquatic invertebrates and water temperature in NCOS?

Methods

- The goal of the study was to collect samples of taxa and water quality at all 12 Deverux Slough sites using pH and YSI meters, as well as 250 m filtered beakers (FB250) and eDNA (Environmental DNA). More specifically, the invertebrate samples are collected quarterly (4x/year) by student interns and volunteers using a filtered bucket and dip net method, with the 3 most common invertebrates found with the filtered beaker 250 micron method being copepods, cladocera, ostracods. At the CCBER Roost Lab, the FB250 samples were inspected under a microscope and sorted by invertebrate taxa. The E-DNA samples are lastly sent to an external lab to be further inspected. According to NCOS researchers Grinstead, Huitema, Norman, et al., "this study aims to reveal… difference(s) in macroinvertebrate species diversity and abundance over time…particularly using it as a model to prompt conservation efforts in other locations."
- The results of monitoring invertebrate abundance serve as a foundation for future exploration of how water quality can be a great indicator of the state of a wetland. Understanding how temperature shapes invertebrate abundance at NCOS is a natural extension of this goal; temperature is a direct force acting on aquatic ecosystems, and the invertebrates that respond to it are the same organisms researchers rely on to read the health of the wetland.
- To examine the relationship between water temperature and invertebrate abundance, temperature was divided into five equal ranges spanning the observed extent of the dataset (10.0–38.0°C), and the distribution of log-transformed abundance within each range was visualized for each taxon.

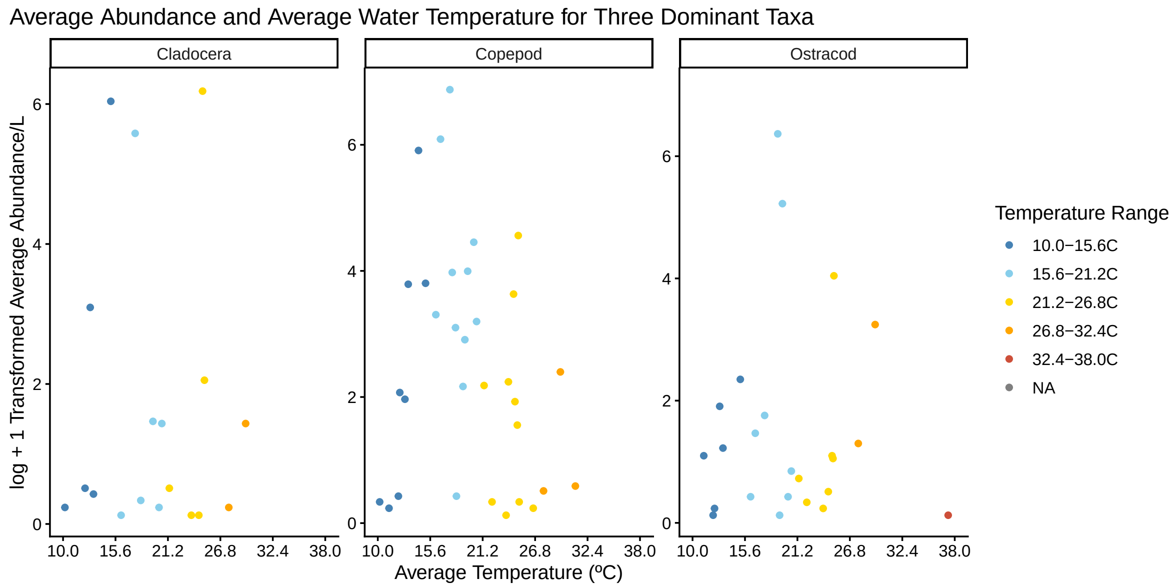
Results

Figure 1 - What is the average abundance distribution of the three dominant taxa?



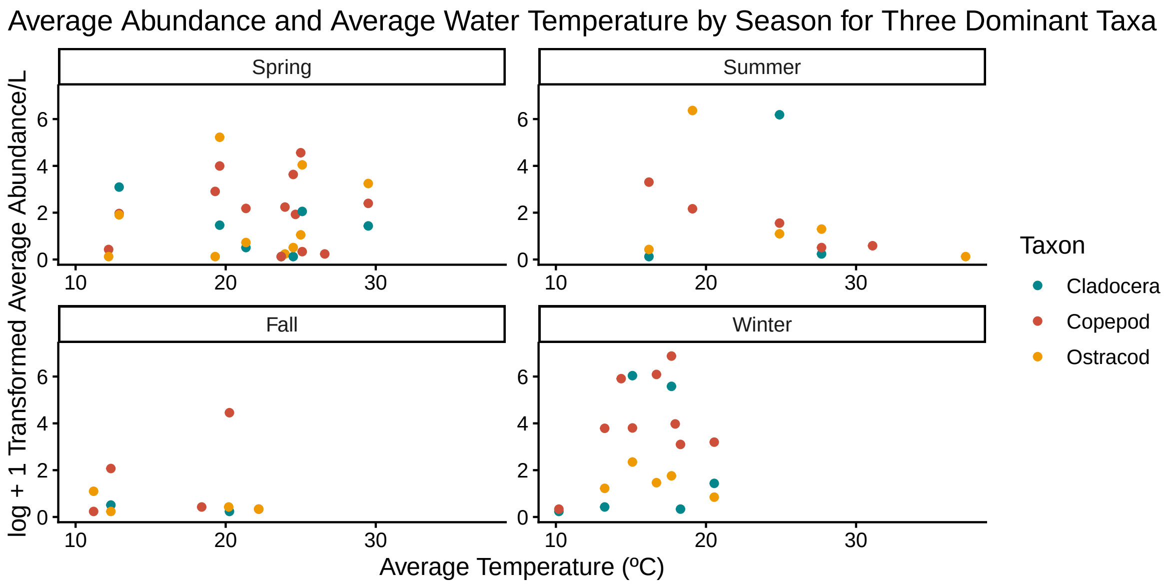
First, we analyzed the variance of average abundance/L across our three taxa in our scope. We found that Copepod was the most abundant at NCOS, followed by Ostracod, then slightly behind by Cladocera. Cladocera seemed to have the widest distribution of average abundance/L, whereas Copepod had the smallest and more consistent distribution across various sample points, and Ostracod had a moderate sized distribution. This figure supports our research question as it visualized specifically ***how*** abundant each taxa is and how spread out their data points were across all sites and temperatures. Our understanding of the variance of abundance on our three taxa was an important precedent on the deeper analysis on water temperature.

Figure 2 - How does abundance vary across temperature ranges for each taxon?



- After understanding specifically ***how*** abundant each taxa was across all sites, we created Figure 2 which displays the relationship between average abundance and average water temperature per sample point Average temperatures varied from 0-38ºC, thus we split up the whole temperature range equally into 5 color coded ranges: dark blue (10.0-15.6ºC), light blue (15.6-21.2ºC), yellow (21.2-26.8ºC), orange (26.8-32.4ºC), and red (32.4-38.0ºC). Cladocera was most abundant in a temperature range between 10.0-21.2ºC, Copepod had the widest abundance range of 10.0-26.8ºC, and Ostracod was most abundant between 15.6º-26.8ºC. This figure supports our research question as it explains directly in which specific range of temperature was there the most average abundance of taxon/L. We can now see the pattern between abundance and water temperature, and can continue diving deeper.

Figure 3 - What is the strength of the relationship between average water temperature and average invertebrate abundance across by season?



- It was clear from Figure 2 that aquatic invertebrates tended to be more abundant from an average water temperature from 10.0-26.8ºC. Thus, we further explored ***when*** exactly in the year this water temperature tended to occur. Figure 3 shows average abundance/L by average water temperature (ºC) for each of the three taxa per season, between 2020-2024. Winter stood out to have the lowest average water temperatures between 10.0-20ºC, thus we concluded that average aquatic invertebrate abundance is most likely highest during the Winter at NCOS. This figure connects to our research question as it directly visualizes the correlation between water temperature and aquatic abundance, but more specifically during four time ranges.

Discussion

Connecting Results to NCOS Reports/Publications

The results align with findings from Stiles & Kracha, whose poster on a restored California wetland documented shifts in invertebrate assemblages as the system matured over three years (Stiles and Kracha 2023). Given that NCOS restoration began in 2017 and sampling commenced in 2018, the data collected likely reflect an ecosystem still establishing ecological equilibrium. The winter abundance spike observed is consistent with Stiles & Kracha's finding that seasonal rainfall reduces salinity stress, driving fluctuations in invertebrate community composition (Stiles and Kracha 2023).

Connecting Results to Other Ecosystems

The temperature-abundance relationship observed at NCOS finds support in studies conducted across comparable wetland systems. Marchetti et al. found that invertebrate abundance, richness, and diversity increased with wetland age across restored Sacramento Valley wetlands, with Ostracoda — one of the three taxa monitored in this study — comprising over 20% of all samples (Marchetti, Garr, and Smith 2010). Given that NCOS is a relatively young restoration site, the abundance patterns documented here likely reflect an early stage of that same successional trajectory. Suren et al. similarly demonstrated that invertebrate communities recolonized successfully following hydrological restoration with minimal intervention, suggesting that restored wetlands across geographic contexts follow comparable recovery dynamics (Suren, Lambert, and Sorrell 2011).

Challenges in Data Collection

Obtaining representative water samples was complicated by particulate matter — including mud, sticks, and organic debris — that required filtration and likely introduced variability in usable sample volume across collection events. Some sampling locations, such as areas beneath bridges or in seasonally waterlogged margins, posed physical access challenges that may have limited consistency across seasons.

Other Variables That Would Influence Observed Patterns

Salinity is the most directly relevant confounding variable, as winter rainfall reduces salinity independently of temperature, meaning some of the abundance patterns observed may partly reflect salinity dynamics rather than thermal effects alone; salinity is typically measured using a conductivity meter or refractometer at the time of collection. Dissolved oxygen (DO) is an additional variable of interest, given that warmer water holds less oxygen and could suppress invertebrate abundance at elevated temperatures independently of direct thermal stress; DO is measured via probe concurrent with sampling. Local vegetation cover and canopy shading may also introduce spatial variability, as shaded microsites can register cooler temperatures independent of seasonal conditions.

Temporal Changes and Dataset Limitations

Seasonal sampling captures broad temporal trends but cannot account for finer-scale environmental variation; a single sample per season represents a snapshot that may not reflect conditions experienced by invertebrate populations over preceding weeks. Since invertebrate community structure responds to accumulated conditions rather than instantaneous measurements, short-term fluctuations between sampling events go undetected. Without continuously deployed temperature loggers at each of the 12 Devereux Slough sites, it remains difficult to assess whether recorded temperatures are representative of the broader inter-sampling period, and multi-year data would be needed to distinguish restoration-driven trajectories from recurring seasonal cycles.

References

Batzer, Darold P., Barbara E. Taylor, Adrienne E. DeBiase, Susan E. Brantley, and Richard Schultheis. 2015. "Response of Aquatic Invertebrates to Ecological Rehabilitation of Southeastern USA Depressional Wetlands." *Wetlands* 35 (4): 803–13. https://doi.org/10.1007/s13157-015-0671-1.

Bonacina, Luca, Federica Fasano, Valeria Mezzanotte, and Riccardo Fornaroli. 2022. "Effects of Water Temperature on Freshwater Macroinvertebrates: A Systematic Review." *Biological Reviews* 98 (1): 191–221. https://doi.org/10.1111/brv.12903.

Dodson, Stanley I., and Richard A. Lillie. 2001. "Zooplankton Communities of Restored Depressional Wetlands in Wisconsin, USA." *Wetlands* 21 (2): 292–300. https://doi.org/10.1672/0277-5212(2001)021%5B0292:zcordw%5D2.0.co;2.

Li, Changling, Xiaoxia Luo, Xianghu Huang, and Binhe Gu. 2009. "Influences of Temperature on Development and Survival, Reproduction and Growth of a Calanoid Copepod (*Pseudodiaptomus Dubia*)." *The Scientific World JOURNAL* 9: 866–79. https://doi.org/10.1100/tsw.2009.96.

Lu, Kangle, Haitao Wu, Qiang Guan, and Xianguo Lu. 2020. "Aquatic Invertebrate Assemblages as Potential Indicators of Restoration Conditions in Wetlands of Northeastern China." *Restoration Ecology* 29 (1). https://doi.org/10.1111/rec.13283.

Marchetti, Michael P., Mariah Garr, and Adam N. H. Smith. 2010. "Evaluating Wetland Restoration Success Using Aquatic Macroinvertebrate Assemblages in the Sacramento Valley, California." *Restoration Ecology* 18: 457–66. https://doi.org/10.1111/j.1526-100X.2008.00468.x.

Stiles, L., and C. Kracha. 2023. "Assessing Aquatic Invertebrate Establishment in a Restored Wetland over Time." UC Santa Barbara: Cheadle Center for Biodiversity; Ecological Restoration. https://escholarship.org/uc/item/00t2k2t7.

Suren, Alastair M., Paul Lambert, and Brian K. Sorrell. 2011. "The Impact of Hydrological Restoration on Benthic Aquatic Invertebrate Communities in a New Zealand Wetland." *Restoration Ecology* 19: 747–57. https://doi.org/10.1111/j.1526-100X.2010.00723.x.